

OpenGeoMetadata API Mirror Network

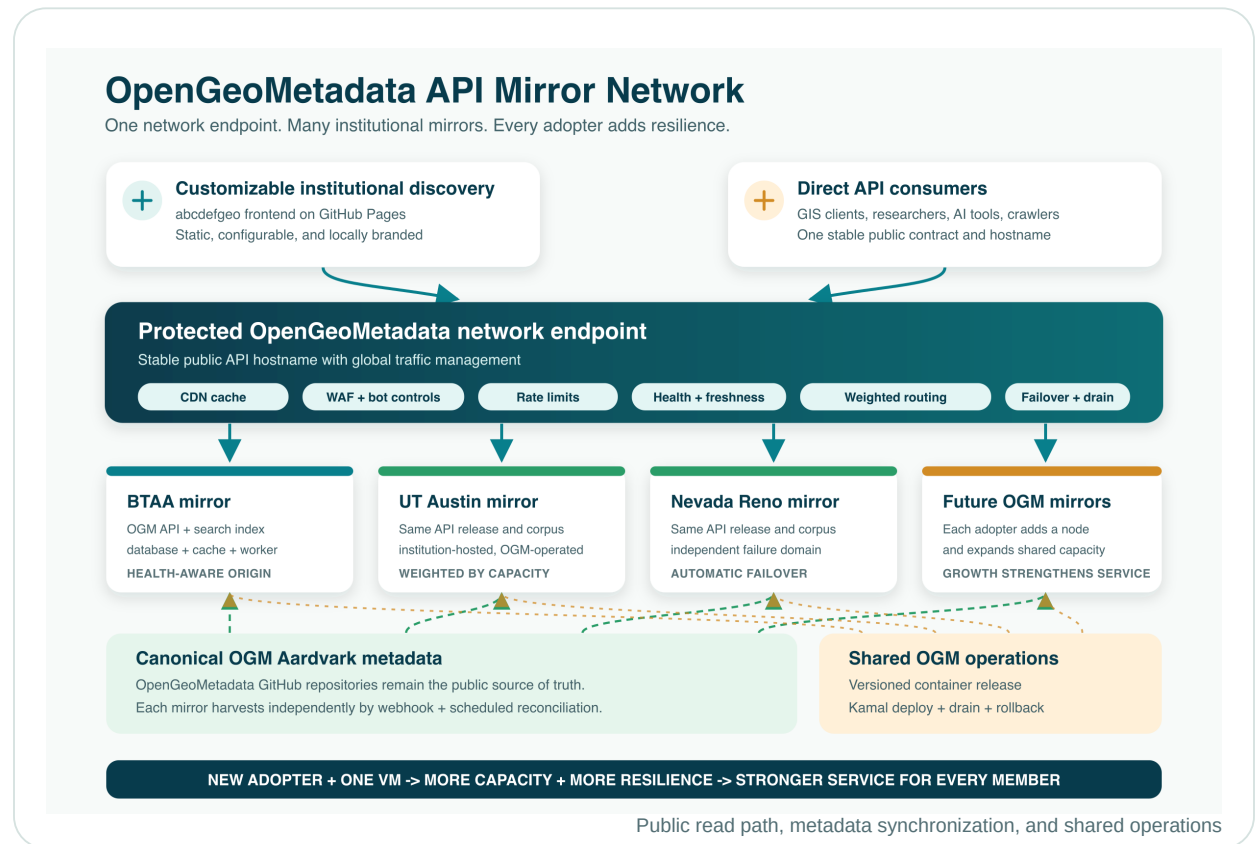
Shared infrastructure that becomes stronger with every new member

EXECUTIVE SUMMARY

OpenGeoMetadata already has the essential building blocks: the OGM Aardvark schema, public GitHub metadata repositories, a proven API, and the configurable **abcdefgeo** frontend for GitHub Pages.

The proposed mirror network adds a protected global endpoint in front of compatible institutional OGM API nodes. Each participating library contributes one ordinary Linux VM. The OGM operator deploys and updates the application with Kamal, synchronizes the public corpus, monitors readiness, and routes traffic by health and capacity.

The strategic advantage is simple: every new adopter gains the foundation for a customizable, institution-branded discovery platform *and* adds capacity, geographic diversity, and resilience for the entire community. A campus can drain its node for upgrades while users continue through the remaining mirrors.



An institution contributes the equivalent of one modest server and gains the foundation for a customizable, institution-branded discovery platform backed by the combined capacity and resilience of the OpenGeoMetadata community.

The smallest possible institutional ask; network-wide return

Institution hosts the machine. OGM operates the application. Every member benefits from the fleet.

COST AND EFFORT

\$1,500 / year

planning target per externally purchased mirror node

Reference: 16 vCPU, 64 GB RAM, 500 GB usable SSD/NVMe, 1 Gbps.
Sized from BTAA production; target 4-8 provisioning hours plus a few host-support hours per quarter.

A campus VM from existing capacity may have little or no incremental cost.

WHY THIS IS INEXPENSIVE

No local software project. No institution-specific backend. No frontend server. No cross-campus database replication. Public Aardvark records in GitHub remain canonical, so indexes and caches are rebuildable.

The closest public hardware benchmark is a 16-thread / 64 GB / 2x512 GB NVMe dedicated server at about \$117 per month. Shared edge and monitoring costs are budgeted once for the network, not once per campus.

INSTITUTION PROVIDES

One production VM
Firewall, DNS, SSH, and OS coordination
One service sponsor + one technical contact

OGM OPERATES

Kamal install, drain, update, and rollback
Harvest, index, cache, and monitor
Health-aware traffic weights + incidents

EVERY MEMBER GAINS

Customizable, institution-branded discovery
Failover during outages + maintenance
More aggregate capacity with every adopter

DECISION REQUESTED

Approve a **90-day, three-node pilot**: the current BTAA node plus two adopters such as UT Austin and the University of Nevada, Reno. Measure onboarding, cost, API compatibility, corpus freshness, traffic distribution, bot protection, and failover. Return with a production governance and service-level proposal.

Planning sources: opengeometadata.org; github.com/opengeometadata; github.com/geobtaa/api; github.com/ewlarson/ogm-api; github.com/ewlarson/abcdefgeo; Hetzner price adjustment (2026-06-15); Cloudflare Load Balancing docs.